

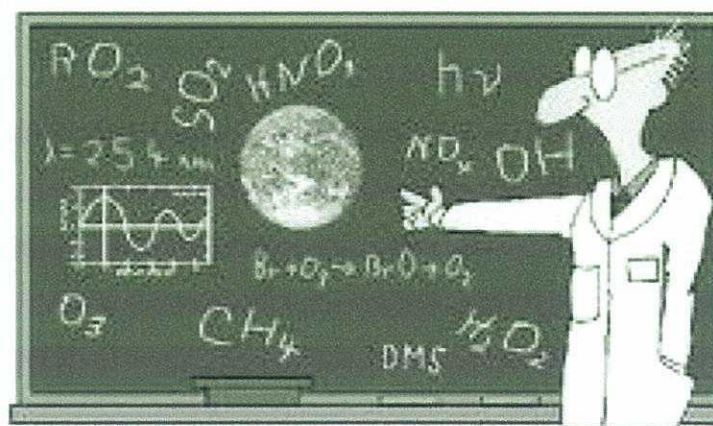
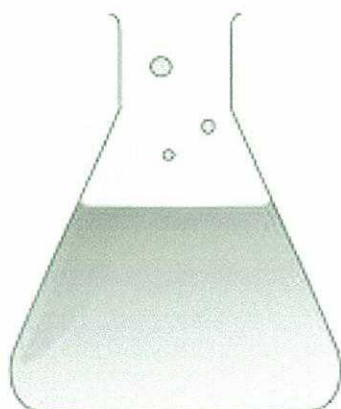
Periodic Table of the Elements

1																	10	11														
3	4																	12	13	14	15	16	17	18								
5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103

* Lanthanide Series
 † Actinide Series

YEAR 10 SCIENCE

CHEMISTRY



YEAR 10 SCIENCE
NATURAL AND PROCESSED MATERIALS
CHEMICAL BONDING - GLOSSARY

Atom - fundamental building block of matter.

Proton - positively charged particle found in the nucleus of an atom.

Neutron - neutral particle found in the nucleus of an atom.

Electron - negatively charged particle found around the nucleus; is electrostatically attracted to the protons in the nucleus.

Element - substance made up of atoms that are all the same (i.e. same atomic number).

Atomic Number - number of protons in the nucleus.

Atomic Mass - the number of protons and neutrons in the nucleus.

Molecule - two or more atoms chemically combined.

Compound - two or more elements chemically combined in fixed proportions.

Periodic Table - arranges elements in order of increasing atomic number but groups them according to their chemical properties.

Period - a row across the periodic table.

Group - a column going down the periodic table; all elements in a group have the same chemical properties.

Metals - found on the left hand side of the periodic table; donate electrons during reactions.

Malleable - able to be beaten into thin sheets.

Ductile - able to be made (drawn) into a wire.

Lustre - the shine of a metal.

Alloy - a mixture of two or more metals.

Metallic Bond - electrostatic attraction between positive metal ions and delocalised electrons.

Non-metals - found on the right hand side of the periodic table; accept electrons during reactions.

Alkali Metals - belong to Group I of the periodic table; react with water to form alkali (basic) solutions.

Alkali Earth Metals - belong to Group II of the periodic table; not as reactive as the alkali metals.

Transition Metals - found in the middle of the periodic table; most reactive metals are at the top.

Halogens - belong to Group VII of the periodic table; form salts when reacted with metals.

Noble (Inert) Gases - belong to Group VIII of the periodic table; do not react with other elements.

Allotrope - a different form of the same element.

Ion - an atom (or group of atoms) that has a charge due to the gain or loss of electrons.

Valency - the charge on an ion.

Ionic Bond - the electrostatic attraction between oppositely charged ions.

Covalent Bond - formed by the sharing of two electrons between two atoms.

Covalent Network - atoms bond to each other in a 3-dimensional arrangement; elements in Group IV do this.

Solute - substance that dissolves.

Solvent - the substance that does the dissolving.

Solution - formed when one substance dissolves into another.

Precipitate - a solid that forms when two ionic solutions are mixed, forming an insoluble substance.

Spectator Ions - not directly involved in forming a precipitate.

Ionic Equation - shows only the ions involved in a reaction.



+

1 Read the following text about scientists and cross out the incorrect verb alternatives. The verbs are underlined. The first one has been done for you. Remember that historical recounts should be written in past tense.

People often ~~made up~~ / make up models to explain / ~~to explained~~ things. Scientists perform / ~~performs~~ experiments which show / ~~shows~~ them how something acts, or what something looks / ~~looked~~ like. This information is used to develop / ~~to develops~~ a model. The model of the atom has / ~~underwent~~ / has undergone many changes.

The idea of atoms is suggested / ~~was suggested~~ by Democritus in ancient Greece (460 – 370 BC). He logically deducted / ~~deduced~~ that matter could not continue to be cut into smaller pieces forever, and that eventually there had to be / ~~have to be~~ a smallest particle, which he calls / ~~called~~ the atom (about 400 BC).

In England, John Dalton (1766 – 1844) heard / ~~heared~~ about the experiments of other chemists. From their results he concluded / ~~concludes~~ that the atoms of the same element was / ~~were~~ identical and cannot be / ~~could not be~~ broken. In a chemical reaction, the atoms were separated, joined or rearranged. Dalton's model of the atom were / ~~was~~ like a ball (1803).

When English scientist Joseph Thomson (1856 – 1940), and other scientists, experiment / ~~experimented~~ with electricity, they finded / ~~found~~ that cathode rays are / ~~were~~ made up of tiny particles which travelled / ~~travel~~ in straight lines and carried / ~~carried~~ a negative electric charge. These cathode rays came / ~~comed~~ from the negative terminal (ie the cathode) and travelled to the positive terminal (ie the anode), when very high voltages have / ~~were~~ applied. Thomson concludes / ~~concluded~~ that the rays make up / ~~were made up of~~ electrons, which were very small (they was / ~~were~~ 2000 times lighter than a hydrogen atom) and had / ~~have~~ a negative charge. But if negative electrons came / ~~come~~ from neutral and relatively large atoms, the atom had to / ~~has to~~ look like a plum pudding: the dough of the pudding was a uniform substance with a positive charge and contained / ~~contain~~ tiny sultanas with a negative charge (1904).

2 In the following text about atoms and radioactivity, there are 12 spelling mistakes. Underline the incorrectly spelt words and write them correctly in the margin.

Radioactivity had been discovered and alpha particles were identified as tiny positive protons. A student of Joseph Thomson, the New Zealand born Ernest Rutherford (1871 – 1937), shot a beam of alpha particals at a very thin sheet of gold foil. He expected the protons to fly straight threw the foil, practically uneffected by the positive and negative particles in the gold atoms. Rutherford was suprised to find that some protons were deflected, some even bounced back to the sauce. Based on his measurements, Rutherford concluded that the atom's protons had to be concentrated in a small nucleus leaving the electrons to suround it. The mass of the atom was determined by the nucleus, which was much heavier than the electrons, and its size was determined by the electrons, which moved around the nucleus. Rutherford's model of an atom looked like a spherical cloud of negative electrons with a positive nucleus in the centre. If the atom was as big as a football feild, the nucleus would be as big as a tennis ball (1911).

The German Max Planck (1858 – 1947) formulated the quantum theory: Energy is given out or taken in by atoms only in lots (called quanta). Niels Bohr from Denmark (1885 – 1962) concluded that electrons jumped from one energy level to another, each jump being worth one qantum of energy. As a result the atomic model became like a soler system: a small nucleus, containing the heavy positive protons, was surrounded by several orbits on which the tiny negative electrons circled the nucleus (1913).

Erwin Schrodinger (1887 – 1961) and Wolfgang Pauli (1900 – 1958), both from Austria, proposed another model for the atom: The electron shells contained subshells of various shapes, each with a maximum of two electrons. But it was impossible to state where a particular electron was at any time; one could only say where an electron was most likley to be. Thus the atom appered like a nucleus inside many electron clouds (1926).

When the English scientist James Chadwick (1891 – 1974) discovered the existance of neutrons, which were as heavy as protons but carried no charge, the atomic model changed to include neutrons inside the nucleus together with the protons (1932).



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3 The following words are snippets from sentences which summarise the text from questions 1 and 2. Sort the snippets from each sentence and then write down the sensible sentences.

a | atom | Democritus from ancient Greece | named | in about 400BC | the smallest particle |.

b | believed | in 1803 | like a ball | that | the atom | The English scientist John Dalton | was |.

c | drew | In 1904 | Joseph Thomson from England | like a positively charged plum pudding with negative plums in it | the atom |.

d | and | concentrated | Ernest Rutherford | In 1911 | in the central nucleus | moved | the negative electrons | onto a cloud surrounding the nucleus | the positive charge |, who was born in New Zealand, |.

e | but | in 1913 | kept | onto several orbits | placed | rather like our solar system | The Danish Niels Bohr | the negative electrons | the positive nucleus |.

f | an atom where the nucleus was surrounded by several orbital clouds | and | combined | developed | in 1926 | The Austrian scientists Erwin Schrodinger and Wolfgang Pauli | the idea of orbits and clouds |.

g | discovered | In 1932 | inside the positive nucleus | neutral neutrons | the English scientist James Chadwick |.

Drawing atomic models

In the paragraphs above various scientists developed certain models of atoms. Each model explained what the scientists knew at that time. New discoveries made new atomic models necessary.

4 Draw each model as described in the text. Next to your diagram write the description of the model as well as the name of the scientist and the year of discovery.

ELEMENTS AND COMPOUNDS

Definitions

Element: substance that contains atoms that are all the same.

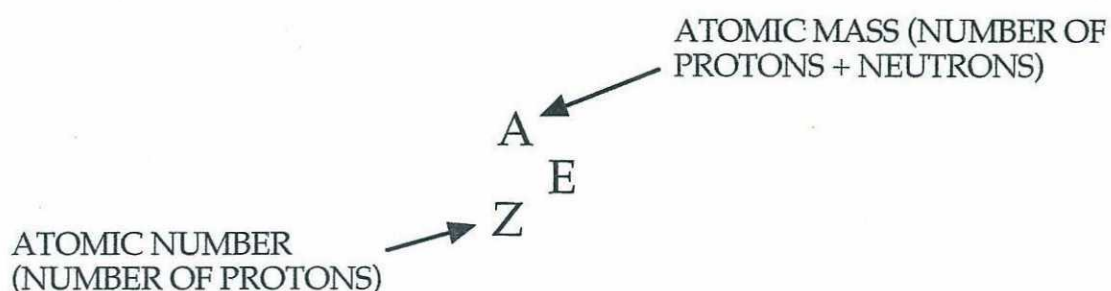
(Elements have atoms with the same number of protons in the nucleus.)

Atomic Number: number of protons in the nucleus.

Mass Number: number of protons and neutrons in the nucleus.

Each element has its own particular number of protons, neutrons and electrons.

These numbers are represented using the following general symbol.



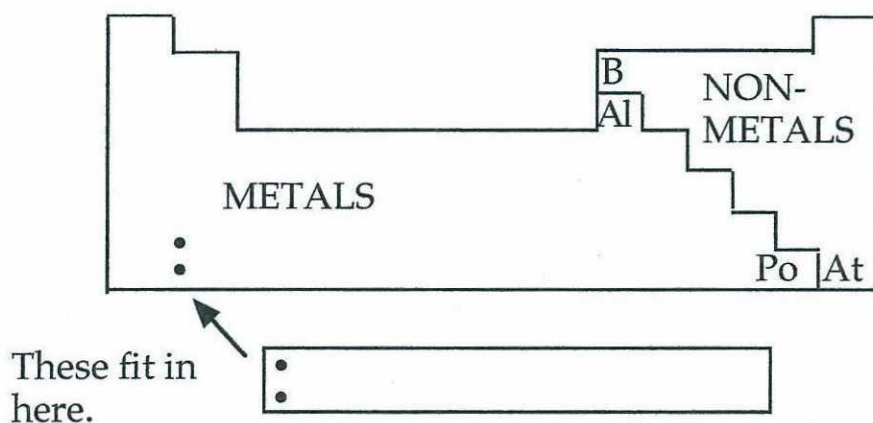
Examples

ELEMENT	PROTONS	NEUTRONS	ELECTRONS
$^{24}_{12}\text{Mg}$	12	12	12
$^{23}_{11}\text{Na}$	11	12	11
$^{238}_{92}\text{U}$	92	146	92

There are 92 naturally-occurring elements and 22 man-made ones.

These are arranged by atomic number in the Periodic Table.

The elements can be classified as metals and non-metals.



Metals are found on the left hand side of the table, non-metals to the right.

Along the diagonal shown are *metalloids* or *semi-metals* that show properties of both metals and non-metals.

The symbols for elements always start with a capital letter, and the second letter (if used) is always lower case.

e.g. Al - aluminium C - carbon

The symbol is based on the English name, though some are based on Latin, Arabic and German names.

NON-ENGLISH

sodium	Na
potassium	K
iron	Fe
copper	Cu
silver	Ag
tungsten	W
gold	Au
mercury	Hg
lead	Pb
tin	Sn

cobalt	Co
manganese	Mn
bromine	Br
iodine	I
zinc	Zn
nickel	Ni
barium	Ba

ENGLISH

hydrogen	H
helium	He
lithium	Li
beryllium	Be
boron	B
carbon	C
nitrogen	N
oxygen	O
fluorine	F
neon	Ne
magnesium	Mg
aluminium	Al
silicon	Si
phosphorus	P
sulfur	S
chlorine	Cl
argon	Ar
calcium	Ca

PERIODIC TABLE

1 H 1.01	2 He 4.0																
3 Li 6.94	4 Be 9.01	5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2										
11 Na 23.0	12 Mg 24.3	13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 39.9										
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc 98	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71 See below	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po 209	85 At 210	86 Rn 222
87 Fr 223	88 Ra 226	89 Ac 227.0	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np 237.0	94 Pu 244	95 Am 243	96 Cm 247	97 Bk 247	98 Cf 251	99 Es 252	100 Fm 257	101 Md 258	102 No 259		

Table of Salt Solubilities

SOLUBILITY SERIES

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb
138.9	140.1	140.9	144.2	145	150.4	152.0	157.2	158.9	162.5	164.9	167.3	168.9	173.0
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No
227.0	232.0	231.0	238.0	237.0	244	243	247	247	251	252	257	258	259

6	Atomic Number
C	Symbol
12.0	Relative Atomic Mass (Atomic Weight)

RULE	EXCEPTION
ALL SALTS of K^+ , Na^+ and NH_4^+ are SOLUBLE	
ALL NITRATE compounds are SOLUBLE	
ALL ACETATE compounds are SOLUBLE	
ALL CHLORIDE compounds are SOLUBLE	$AgCl$, $CuCl$, $PbCl_2$
ALL SULPHATE compounds are SOLUBLE	Ag_2SO_4 , $BaSO_4$, $CaSO_4$, $PbSO_4$
ALL SULPHIDE compounds are INSOLUBLE	BaS , CaS , K_2S , MgS , Na_2S , $(NH_4)_2S$
ALL CARBONATE compounds are INSOLUBLE	K_2CO_3 , Na_2CO_3 , $(NH_4)_2CO_3$
ALL PHOSPHATE compounds are INSOLUBLE	K_3PO_4 , Na_3PO_4 , $(NH_4)_3PO_4$
ALL OXIDES, HYDROXIDES are INSOLUBLE	K^+ , Na^+ etc.

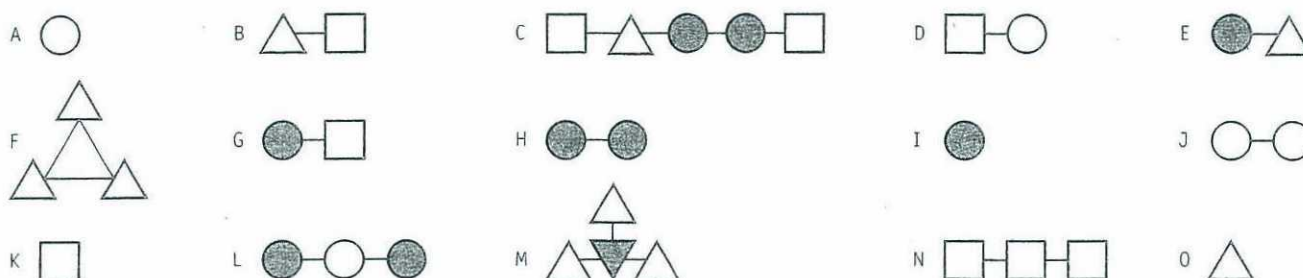
K Na Ca Mg Al Zn Fe Ni S S O H
K Na Ca Mg Al Zn Fe Ni S S O H

Matter can be divided into groups using certain criteria. Matter can be divided into living and non-living things. Matter can be divided into solids, liquids and gases. Matter can be grouped into metals and non-metals. Matter can be split into single particles and multiple particles. Matter can be divided into substances which contain one kind of atom or more than one kind of atom.

Definitions

- **Atoms** are tiny particles. They are the simplest form of matter. Atoms are made up of single particles.
- **Molecules** are also tiny particles. They are made up of two or more atoms. These atoms are held together by chemical bonds.
- **Elements** are substances which contain only one kind of atom.
- **Compounds** are substances which contain particles that are made up of two or more different atoms.

Look at the following diagrams showing various particles and substances. Then answer the questions below by referring to the definitions.



1 Identify all atoms and all molecules and record them in the table.

Particles	Atoms	Molecules
Diagrams		

2 Identify all elements and all compounds and record them in the table.

Substances	Elements	Compounds
Diagrams		

3 Complete the following table about particles and substances by adding as many examples of each as possible from the diagrams.

Atom of element	Atom of compound	Molecule of element	Molecule of compound

4 Explain why one cell in the table from question 3 is left empty. _____

5 Make up two more examples for each cell in question 3 and draw them below the table.

6 Explain what the lines in the diagrams represent. _____



5

Periodic Table

The Periodic Table arranges the elements according to their chemical properties.

Metals are found to the left hand side of the zig-zag on the table. **Non-metals** are found to the right.

Elements on either side of the zig-zag are called **semi-metals** or **metalloids**, showing characteristics of both metals and non-metals.

Hydrogen is a non-metal that behaves like a Group I element sometimes and a Group VII at other times. It is sometimes shown by itself in the table.

All metals with the same chemical properties are arranged in **groups** (vertical columns). They are numbered from I - VIII across the table.

The horizontal rows are called **periods**.

I		II																				III	IV	V	VI	VII	VIII							
1 H 1.008																																		2 He 4.003
3 Li 6.941	4 Be 9.012																			5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18									
11 Na 22.99	12 Mg 24.30	Transition metals																		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95									
19 K 39.10	20 Ca 40.08																			21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3																	
55 Cs 132.9	56 Ba 137.3	57 *La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)																	
87 Fr (223)	88 Ra (226)	89 §Ac (227)	104 Unq (261)	105 Unp (262)	106 Unh (263)	107 Uns (262)	108 Uno (265)	109 Une (266)	110	111																								
* Lanthanide series			58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.2	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0																		
§ Actinide series			90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)																		
6 C 12.01	← Atomic number																																	
	← Symbol																																	
	← Relative atomic mass (atomic weight)																																	

A relative atomic mass in brackets is the mass number of the isotope with the longest half-life

Trends Across the Table

Consider the row 3 elements.

- Go from metals → semi-metals → non-metals.
- Reactivity (ability to react) decreases.

- Conductivity decreases.

i.e. Metals conduct electricity well - non-metals are insulators.

- Melting and boiling points increase from Group I - IV, then decrease from Group V - VIII.

The stronger the bonding between particles, the higher the melting and boiling point.

e.g. H_2O melts at $0\text{ }^{\circ}\text{C}$, NaCl melts at $801\text{ }^{\circ}\text{C}$. NaCl has the stronger force between its particles.

Trends Down a Group

- Reactivity increases down a metal group (I-III) - decreases down a non-metal group (IV-VIII).

e.g. Fr (francium) is the most reactive metal (bottom LHS of the table).
F is the most reactive non-metal (top RHS).

- Melting and boiling point decreases down metallic groups (I-III) - increases down non-metal groups.

Special Names of Elements

Alkali Metals

- Group I
- React with H_2O to form alkaline or basic solutions.

Alkaline Earth Metals

- Group II.

Halogens

- Group VII.

Inert (Noble) Gases

- Group VIII.
- Unreactive.
- Exist as single atoms.

Transition Metals

- Found in the middle of the table.
- Elements near each other have similar properties.
- Tend to form coloured compounds with non-metals.

TRENDS OF THE PERIODIC TABLE WEB SEARCH

Purpose To investigate some of the characteristics of the periodic table

Use a general search engine such as Dogpile or Metacrawler to find suitable websites that might provide information to answer the questions below. Some useful websites have been listed.

SEARCH: Metacrawler (www.metacrawler.com)

Phrases: (trends of the periodic table), (interactive periodic table)

<http://www.lyncburg.net/hhs/chemistry/trends/>

Click on Cool Periodic Table link.

Quiz on the Table

<http://www.chemcool.com/regents/periodictable/periodictable.htm>

Interactive Table

<http://www.chemicalelements.com/>

<http://www.chemcool.com/>

<http://www.webelements.com/>

<http://www.webelements.com/webelements/scholar/>

<http://www.schoolscience.co.uk/periodictable.html>

Atomic Spectrum

<http://www.colorado.edu/physics/2000/applets/a2.html>

Questions

1. Write down *3 elements (symbols will do)* that belong to the following from the periodic table.
 - (a) Transition metals
 - (b) Alkali metals
 - (c) Non-metals
 - (d) Noble or inert gases
 - (e) Metalloids or semi-metals
 - (f) Halogens
 - (g) Metals
 - (h) Alkali earths
 - (i) Rare earths

2. Define the following as found in the periodic table.

(a) Group

(b) Period

3. Why are elements placed into the same group in the periodic table?

4. The table below shows the third period elements of the periodic table. Fill in the data required.

	Na	Mg	Al	Si	P	S	Cl	Ar
State of the element at 20 °C.								
Melting point								
Atomic radius								
Electrical conductivity								
Reactivity								

Describe the trends you see.

(a) State at 20 °C.

(b) Melting point.

(c) Atomic radius

(d) Conductivity

(e) Reactivity

5. The table below shows Group II elements of the periodic table - *they are metals*. Fill in the data required.

	Be	Mg	Ca	Sr	Ba	Ra
State of the element at 20 °C.						
Melting point						
Atomic radius						
Electrical conductivity						
Reactivity						

Describe the trends you see.

- (a) State at 20 °C.
- (b) Melting point.
- (c) Atomic radius
- (d) Conductivity
- (e) Reactivity

6. The table below shows Group VII elements of the periodic table - *they are non-metals*. Fill in the data required.

	F	Cl	Br	I	At
State of the element at 20 °C.					
Melting point					
Atomic radius					
Electrical conductivity					
Reactivity					

Describe the trends you see.

- (a) State at 20 °C.
- (b) Melting point.
- (c) Atomic radius
- (d) Conductivity
- (e) Reactivity

ATOMIC AND MASS NUMBERS

Write in the charge on a

- (i) proton _____
- (ii) neutron _____
- (iii) electron _____

Do you know the charge law?

Like charges _____

Unlike charges _____

Write down the name of the relationship (*attraction, repulsion*) between the following particles

proton and proton _____

proton and electron _____

proton and neutron _____

electron and electron _____

electron and neutron _____

The Atomic Number (Z) is the number of _____ in the _____
 The Mass Number (A) is the number of _____ in the _____

Complete the following table

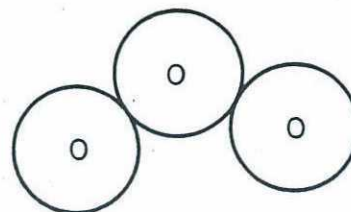
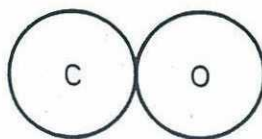
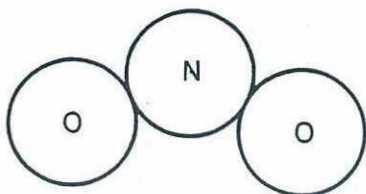
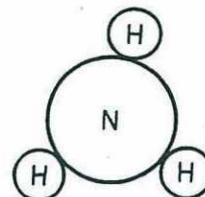
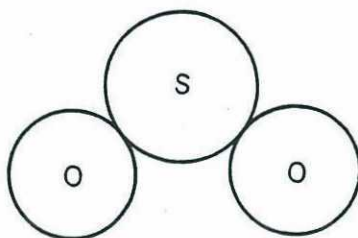
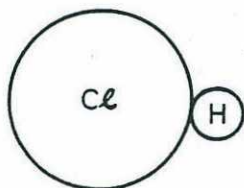
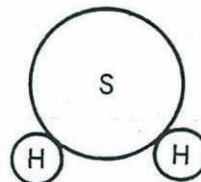
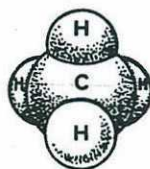
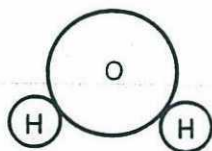
SUBSTANCE	NAME	MASS NUMBER (A)	ATOMIC NUMBER (Z) (No. OF PROTONS)	No. OF NEUTRONS (N)
$^{12}_6\text{C}$				
N		14		7
O			8	8
P		31	15	
$^{35}_{17}\text{Cl}$				
Ca		40		20
Fe				
Ag		108	47	

NAME THE COMPOUND

1. Complete this table.

FORMULA OF SUBSTANCE	ELEMENTS CONTAINED IN SUBSTANCE	NAME OF SUBSTANCE
CaCO_3	calcium, carbon, oxygen	calcium carbonate
CO_2		
Al_2O_3		
KI		
Fe_2O_3		
SO_2		
NaCl		
PbSO_4		

2. Examine these models of compounds and write the formula of each compound.



ELEMENTS, COMPOUNDS AND MIXTURES

1. Elements are pure substances made up of only one kind of _____. They cannot be broken down into simpler substances.

2. There are _____ naturally occurring elements and each has its own chemical symbol.

3. Write symbols for these elements:

aluminium _____	calcium _____	carbon _____	chlorine _____
copper _____	gold _____	hydrogen _____	iron _____
lead _____	magnesium _____	manganese _____	mercury _____
nickel _____	nitrogen _____	oxygen _____	silicon _____
silver _____	sodium _____	sulfur _____	zinc _____

4. Elements can react together chemically to form c _____.

5. If substances are stirred or mixed together, without a chemical reaction taking place between them, then a m _____ is produced.

6. Indicate if the substances below are elements, compounds, or mixtures:

iron _____	carbon dioxide _____
air _____	water _____
salt solution _____	iron and sulfur _____
oxygen _____	sulfur _____

7. Chemists make use of prefixes in naming compounds. What do these prefixes mean?

mono _____, di _____, tri _____, tetra _____, penta _____

8. Write chemical formulas for these compounds:

carbon monoxide _____	sulfur dioxide _____
sulfur trioxide _____	carbon tetrachloride _____
phosphorus pentachloride _____	dinitrogen tetroxide _____

9. Complete the table below:

FORMULA	NAME OF COMPOUND	NUMBER AND TYPE OF ATOM
CO ₂		
NO ₂		
NaCl		
CS ₂		
HCl		
CuSO ₄		
CaCO ₃		

10. Write a word equation for the reaction between

A. iron and sulfur to form iron sulfide

B. magnesium and oxygen to form magnesium oxide

C. hydrogen and oxygen to form water

YEAR 10 BIOLOGICAL CHEMISTRY
WORKSHEET: PERIODIC TABLE, COMPOUNDS

1. Define the following.

- (a) element
- (b) compound
- (c) mixture

2. Write the symbols for the following elements. (Try not to look!)

- | | | |
|--------------|---------------|---------------|
| (a) chlorine | (e) copper | (i) argon |
| (b) sodium | (f) tungsten | (j) manganese |
| (c) iodine | (g) potassium | (k) boron |
| (d) calcium | (h) silver | (l) lead |

3. Classify the elements in Question 2 as metal or non-metal in the table below.

METALS

NON-METALS

4. Name the following compounds.

- | | |
|----------------------------------|-----------------------------|
| (a) $\text{Al}_2(\text{CO}_3)_3$ | (d) NaOH |
| (b) PbSO_4 | (e) SO_3 |
| (c) CO_2 | (f) H_2SO_4 |

5. For each of the compounds in Question 4, write down the name and number of each type of element.

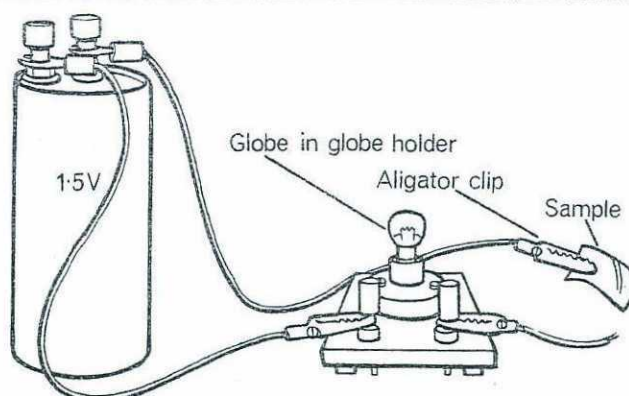
- | | |
|-----|-----|
| (a) | (d) |
| (b) | (e) |
| (c) | (f) |

6. What is the distinction between an organic compound and an inorganic compound?

HOW WELL DO METALS CONDUCT ELECTRICITY?

You will need:

metal samples
two non-metal samples
and this circuit



METHOD

1. Construct the circuit shown above.

2. Choose one of your metal samples.

3. Place the alligator clips on your sample, one at each end.

4. Does the light light up?

5. Repeat the procedure for the remaining samples.

RECORD YOUR OBSERVATIONS

SAMPLE	OBSERVATIONS

QUESTIONS TO CONSIDER

• Which samples conducted electricity? Explain your answer.

- * Consider an electric toaster as an example. Describe the materials used and the reason for their use in the following:

electrical cord _____

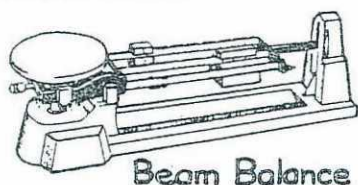
element _____

handles _____

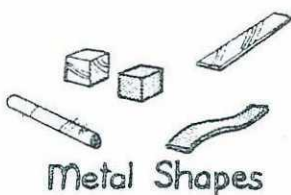
element support _____

DENSITY OF METALS

You will need:



Beam Balance



Metal Shapes

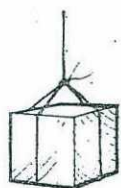


Thread

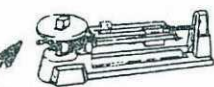


Cotton wool
Graduated cylinder

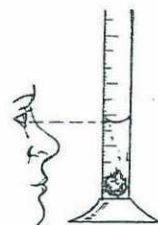
Method:



Tie a piece of cotton thread around a metal sample



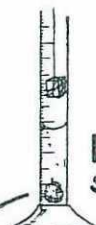
Weigh the sample record its mass



Place same cotton wool at the bottom of the cylinder. Half fill with water and measure the volume.



Record the new level of water and calculate the volume of water displaced



Lower the sample into the water

Repeat the same procedure with each of the metal samples

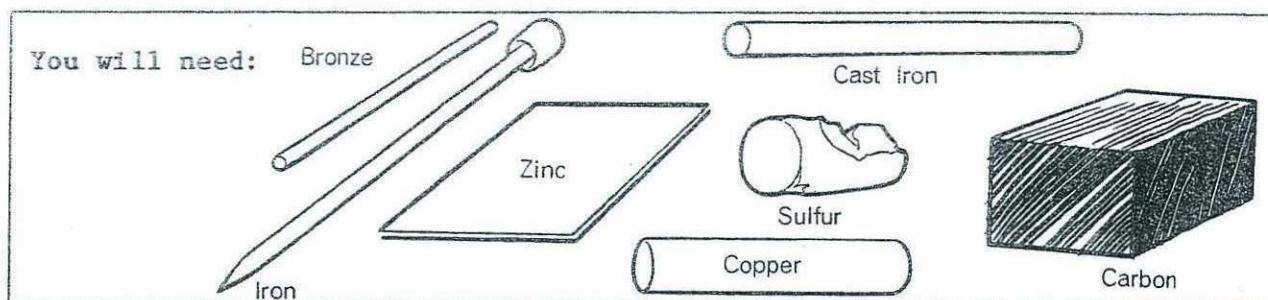
RESULTS

Sample	Mass	Volume water (V_1)	Volume water + (V_2) sample	Volume ($V_2 - V_1$) sample	Density
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

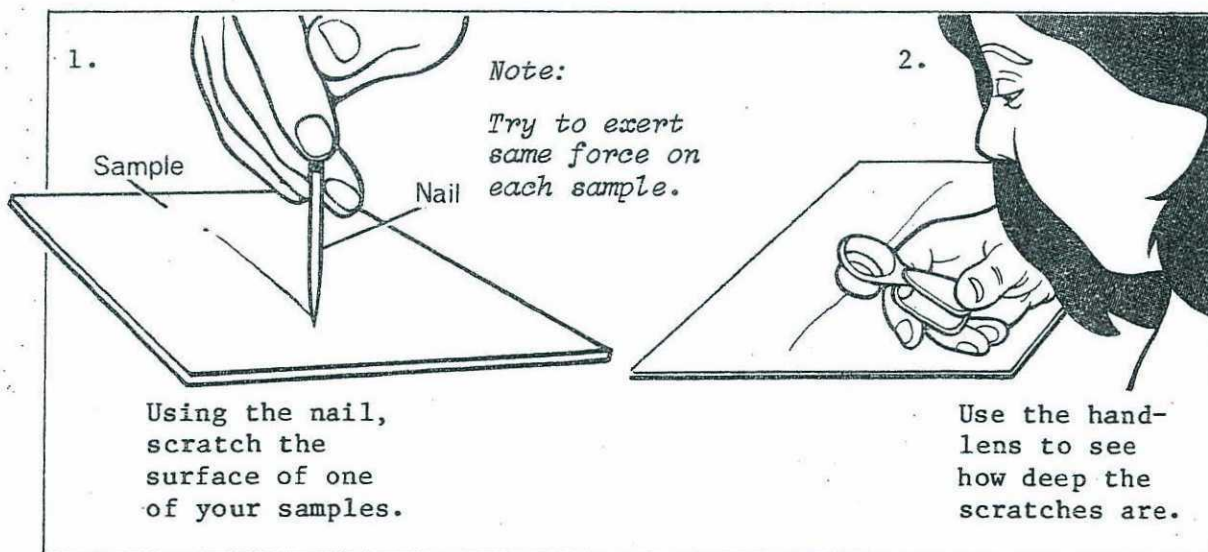
CALCULATE THE DENSITY USING THE FORMULA $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$

Rank your metal samples from least dense to most dense. Consider the uses of these metals in relation to their densities. (Mercury, for example is used in barometers because it is very dense).

HARDNESS OF METALS



METHOD



3. Rank the hardness of your samples

CONSIDER THESE QUESTIONS

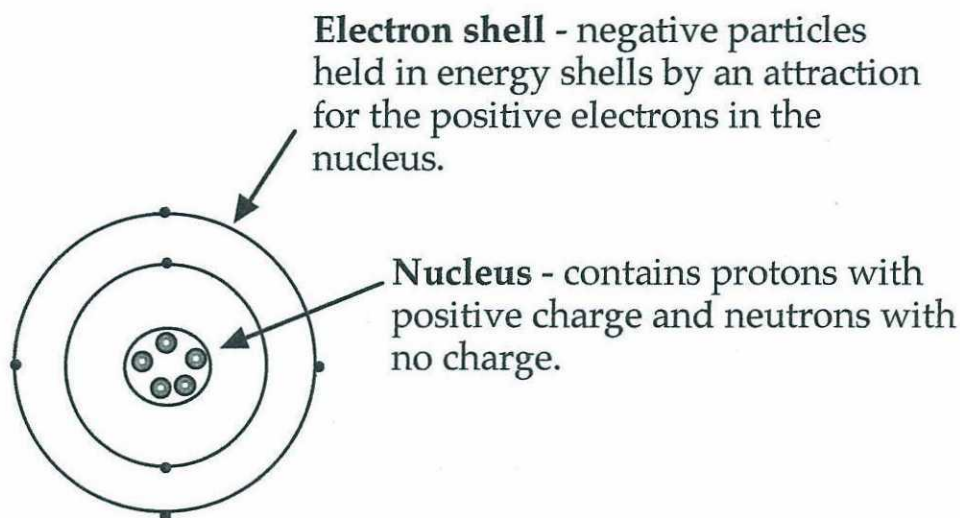
- Which of the metals tested appear as though they could easily be rolled into sheets? (That is they are relatively soft.)
- Where are sheets of these metals used?
- Which metals appear as though they may have to be heated, before they can be re-shaped? (That is they are relatively hard and brittle.)
- Where are these metals most useful?
- What happens if cast iron is bent?
- Which metals could easily be made into new shapes by hammering? (That is, by pressure only.)

EXTRAS FOR EXPERTS

- Which metals would you expect to be the most brittle?
- Why do you think different metals have different hardness and strength?

ATOMIC STRUCTURE

Bohr Model of the Atom



Element - contains only one type of atom.

Compound - two or more elements chemically combined in fixed proportions.

Molecule - two or more atoms chemically combined.

Energy Shells - electrons are found in distinct energy shells around the nucleus.

Each of the energy shells has a maximum number of electrons that can be held.

WE WILL CONSIDER THE FIRST 20 ELEMENTS ONLY.

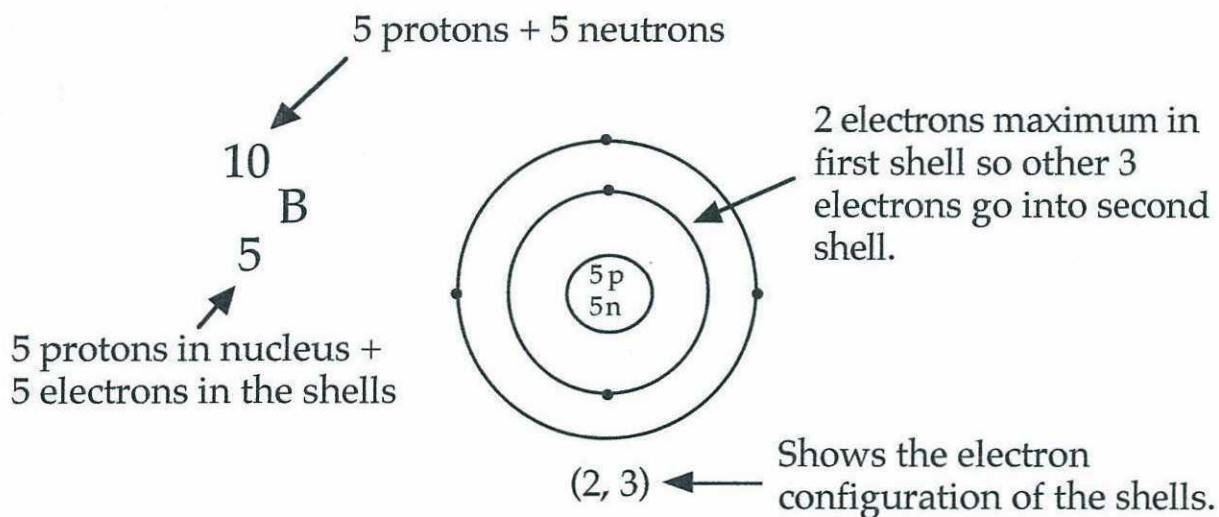
1 st shell	2
2 nd shell	8
3 rd shell	8
4 th shell	2

The arrangement of electrons in the shells is called the *electron configuration* for the element.

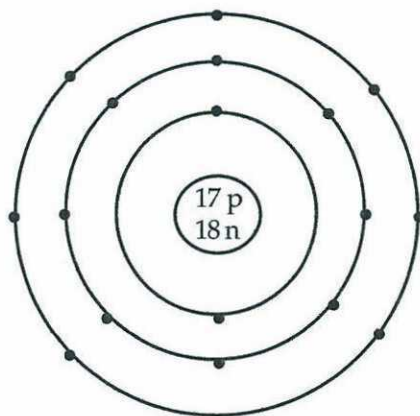
It is possible to draw diagrams to show the electron configuration of elements.

The Periodic Table is very handy for this!

Examples



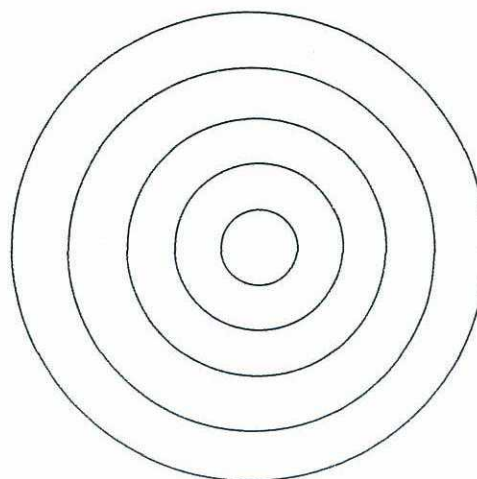
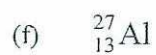
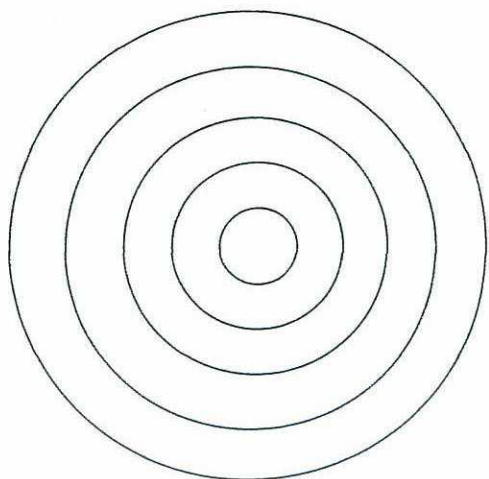
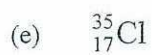
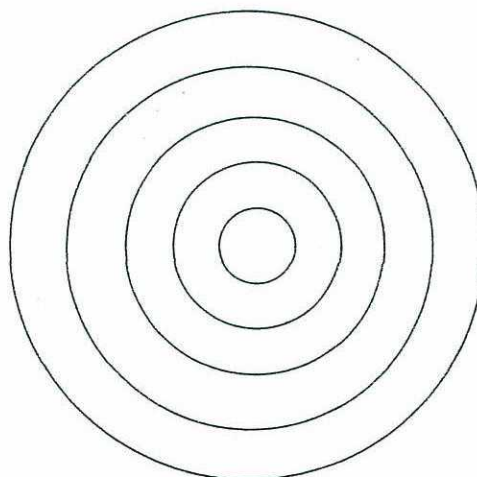
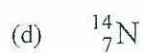
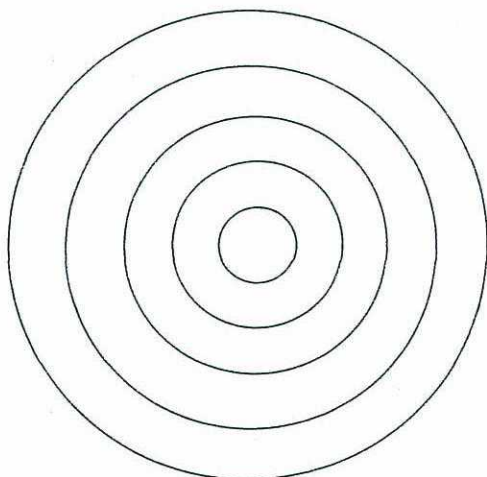
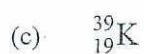
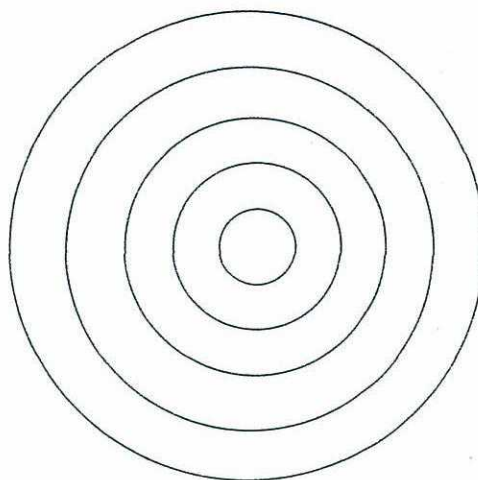
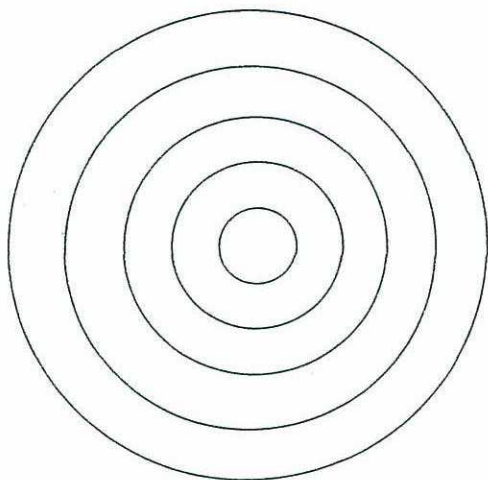
35
Cl
17



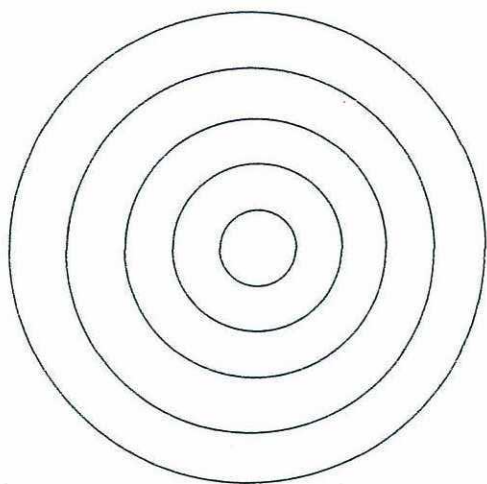
(2, 8, 7)

YEAR 9 SCIENCE
NATURAL AND PROCESSED MATERIALS
WORKSHEET - ELECTRON CONFIGURATION DRAWINGS

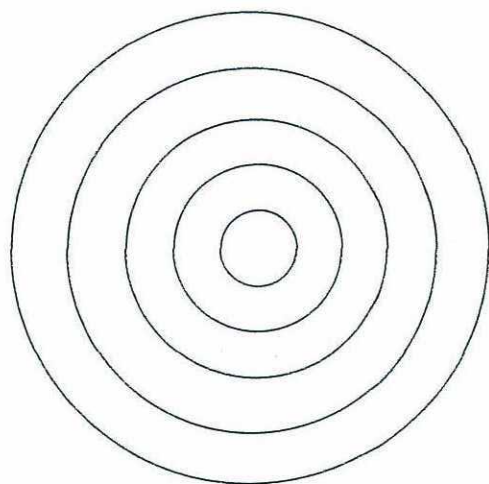
Draw electron configuration diagrams for the following elements. Show the *number of protons and neutrons* in the nucleus, and the *number of electrons* in each shell.



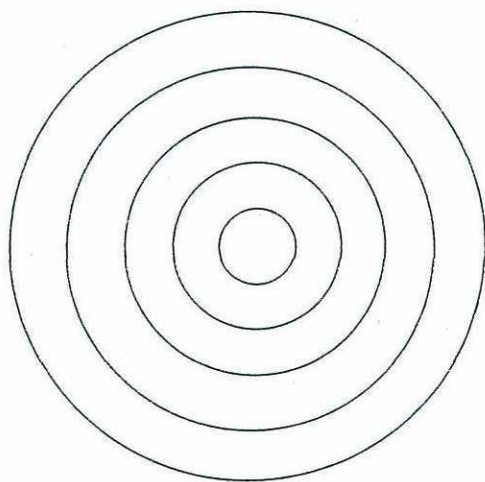
(g) $^{12}_6\text{C}$



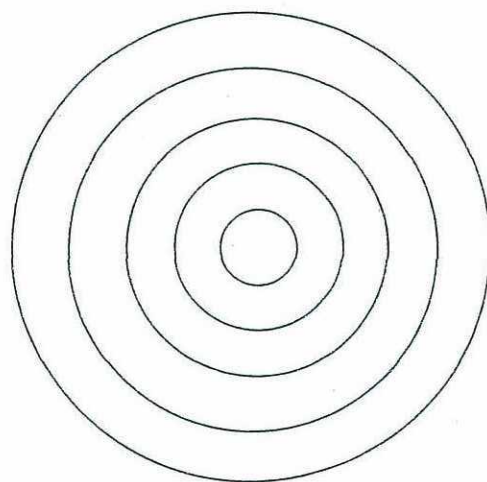
(h) ^4_2He



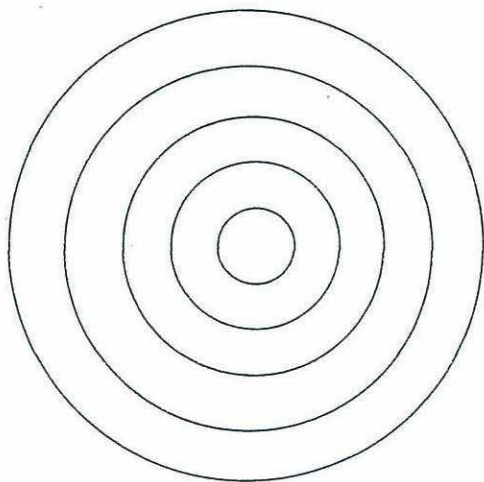
(i) $^{32}_{16}\text{S}$



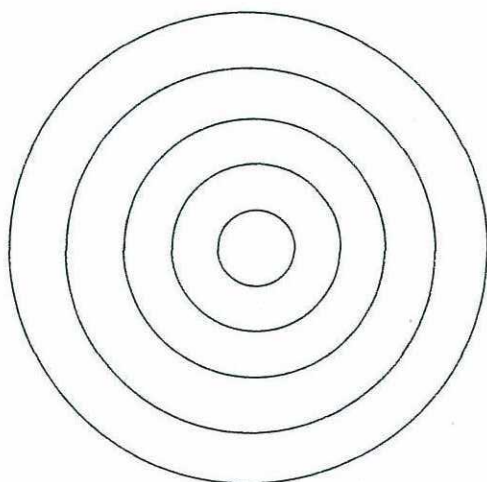
(j) $^{31}_{15}\text{P}$



(k) $^{10}_5\text{B}$



(l) $^{40}_{20}\text{Ca}$

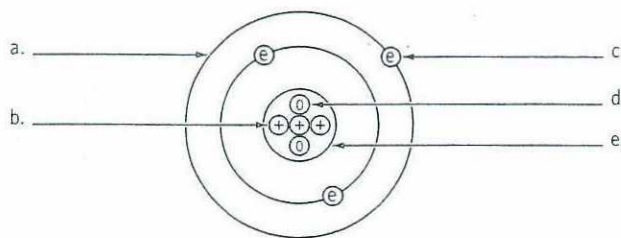


Note: You will need to refer to a copy of the periodic table for this worksheet.

A popular atomic model

Many people today believe that atoms can be described by the atomic model of the solar system atom. This orbital atomic model allows us to explain many processes.

The atom is made up of three main subatomic particles which are grouped into two lots. Protons and neutrons are tightly packed into the nucleus which forms the centre of the atom. Electrons surround the nucleus on several orbits (or shells).



1 Label the diagram of an atomic model (lithium).

Subatomic particles

Protons and neutrons are 2000 times heavier than electrons. Protons and electrons have an electric charge: protons are positive and electrons are negative. Electrons fly around the nucleus on orbits. Neutrons and protons rest in the nucleus.

2 Complete the table about subatomic particles.

Name	Charge	Mass	Position	Movement
			on orbits	
		2000 times		
	neutral			no

Atomic numbers and mass numbers, atoms and isotopes

Atoms are neutral, although some of their subatomic particles do have an electric charge. But the positive and negative subatomic charges cancel each other leaving the atom neutral. Therefore, the number of positive protons must equal the number of negative electrons. The *atomic number* gives the number of protons.

Since protons and neutrons are very much heavier than electrons, it is the protons and neutrons which determine the mass of the atoms. Usually an atom has as many neutrons as protons. The *mass number* is the sum of protons and neutrons.

Some elements have atoms which have the same number of protons in the nucleus (so the same atomic number) but more or fewer neutrons (so they have different atomic weights). Their atoms are now called *isotopes*.

When the number of neutrons is much larger than the number of protons in a large nucleus, the isotope's nucleus is unstable. As a result it breaks up, forming new atoms and giving off radiation.

3 Complete the table about mass and atomic numbers of atoms.

Name	Mass number	Atomic number	Number of protons	Number of neutrons	Number of electrons	Scientific notation
lithium	5	3	3	2	3	${}^5_3\text{Li}$
magnesium	24				12	
aluminium		13		13		
hydrogen	1					
chlorine						${}^{35}_{17}\text{Cl}$
phosphorus						${}^{31}_{15}\text{P}$
oxygen						${}^{16}_8\text{O}$

Electrons on shells

Electrons move on orbits (or shells) around the nucleus. The inside shells are smaller than the outer shells and so can hold fewer electrons. Inner shells are filled up before the outer shells. The first innermost shell can carry a maximum of two electrons, the second and third shells can hold no more than eight electrons (these maximum numbers are true for the first 20 atoms).

4 Draw atomic models for the following atoms. The mass and atomic numbers give you a clue to the number of protons, neutrons and electrons.

helium (${}^4_2\text{He}$)

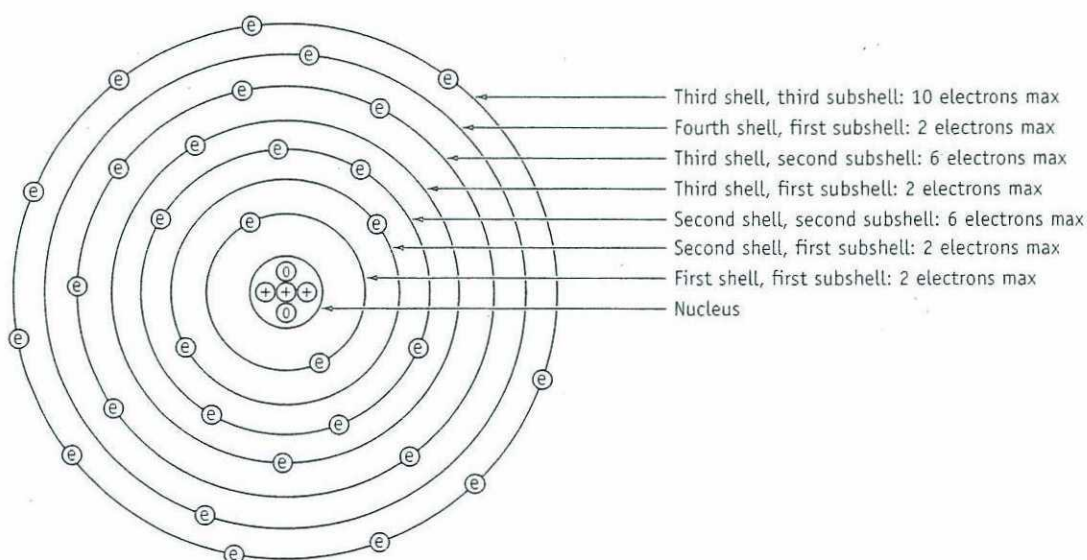
silicon (${}^{28}_{14}\text{Si}$)

beryllium (${}^9_4\text{Be}$)

potassium (${}^{39}_{19}\text{K}$)

Shells and their subshells

The orbital model of the atom becomes untidy with the twenty-first atom, because the electron shells are actually composed of up to four subshells. And some of the subshells overlap each other. The third subshell of the third shell is actually bigger than the first subshell of the fourth shell.



5 State the number of protons, neutrons and electrons for the atom of iron (${}^{56}_{26}\text{Fe}$). _____

6 Shade in electrons to the atomic model above for iron. _____

7 State the number of subshells for the first three shells. _____

FORMATION OF IONS

Definitions

Ion: an atom that has a charge due to the gain or loss of electrons.

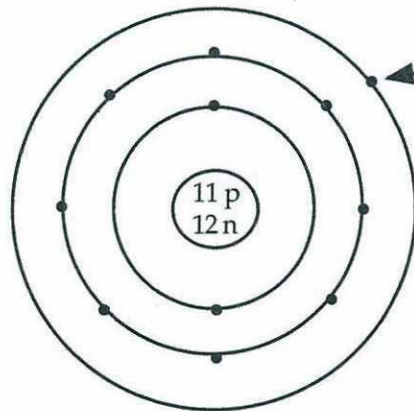
Valency: the charge on an ion.

Positive Ions

- Formed when atoms ***lose electrons***.
- Metal atoms tend to lose electrons from their outer shell.
- Gain a positive charge since there are more protons than electrons.

e.g.

$^{23}_{11}\text{Na}$



Sodium has one electron in its outer shell. If this electron is "lost", the resulting sodium ion will have a +1 charge.

(2, 8, 1)

When the sodium ion forms, it has a full outer shell of eight electrons like the nearest inert gas - neon - which is very stable.

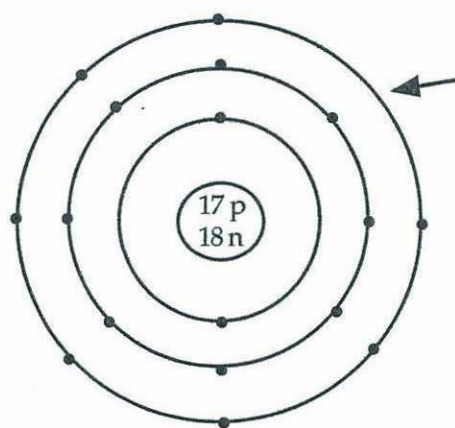
The symbol is written as Na^{1+} .

Negative Ions

- Formed when atoms *gain electrons*.
- Non-metal atoms tend to gain electrons.
- Gain a negative charge since there are more electrons than protons.

e.g.

$^{35}_{17}\text{Cl}$



Chlorine has a space for one electron in its outer shell. If an electron is "gained", the resulting chlorine ion will have a -1 charge.

(2, 8, 7)

By gaining one electron, the chlorine ion has the same completely full outer shell as the nearest inert gas - argon.

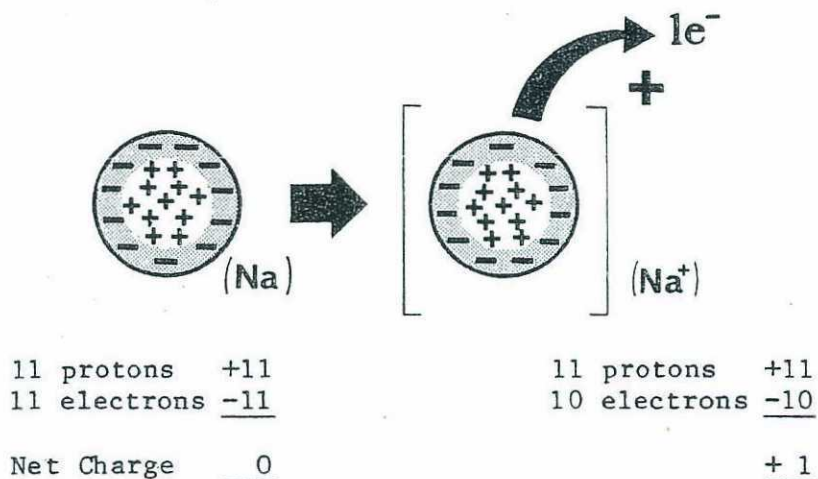
The symbol is written as Cl^{1-} .

You will be given a list of common ions that you can use to name and write the formula of various chemicals.

FORMATION OF IONS

FORMATION OF POSITIVE IONS

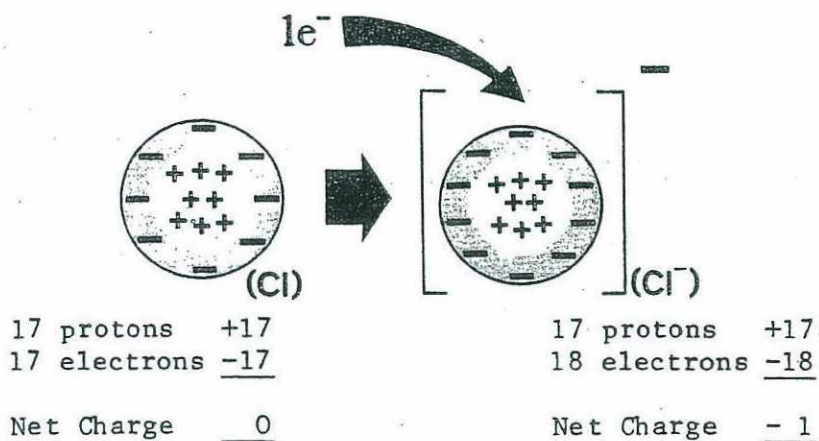
Positive ions are formed when a neutral atom loses *one, two or three electrons* leaving the *ion* with *more protons than electrons*.



CHEMISTS HAVE FOUND THAT METAL ATOMS FORM POSITIVE IONS

FORMATION OF NEGATIVE IONS

Negative ions are formed when a neutral atom gains *one, two or three electrons* leaving the *ion* with *less protons than electrons*.



CHEMISTS HAVE FOUND THAT NON-METAL ATOMS FORM NEGATIVE IONS

NAMES AND FORMULAE FOR SOME COMMON IONS AND POLYATOMIC IONS

+1 charge	-1 charge
lithium, Li^+ sodium, Na^+ potassium, K^+ copper (I) or cuprous, Cu^+ silver, Ag^+	fluoride, F^- chloride, Cl^- bromide, Br^- iodide, I^- hydride, H^-
ammonium, NH_4^+ hydrogen, $\text{H}_{(\text{aq})}^+$	hydroxide, OH^- nitrite, NO_2^- nitrate, NO_3^- acetate, CH_3COO^- hydrogencarbonate or bicarbonate, HCO_3^- hydrogensulfate or bisulfate, HSO_4^- chlorate, ClO_3^-
+2 charge	-2 charge
manganese Mn^{2+} magnesium, Mg^{2+} calcium, Ca^{2+} barium, Ba^{2+} zinc, Zn^{2+} copper (II) or cupric, Cu^{2+} mercury (II) or mercuric, Hg^{2+} iron (II) or ferrous, Fe^{2+} tin (II) or stannous, Sn^{2+} lead (II) or plumbous, Pb^{2+} nickel(II), Ni^{2+}	oxide, O^{2-} sulfide, S^{2-}
	carbonate, CO_3^{2-} sulfate, SO_4^{2-} sulfite, SO_3^{2-}

+3 charge	-3 charge
aluminium, Al^{3+} iron (III) or ferric, Fe^{3+} chromium, Cr^{3+}	nitride, N^{3-}
	phosphate, PO_4^{3-}
+4 charge	
tin (IV) or stannic, Sn^{4+} lead (IV) or plumbic, Pb^{4+}	

THESE NAMES AND FORMULAE MUST BE MEMORISED

WRITING CHEMICAL FORMULAE

Knowledge of the valency of ions allows us to write formulae to represent different chemicals.

“RULES”

1. Write down the ions involved and look at the valency of each.
2. If the numbers are the *same*, take one of each ion.
3. If the numbers are *different*, cross the numbers down to the bottom of each ion.

Examples

magnesium oxide

NUMBERS ARE THE SAME.

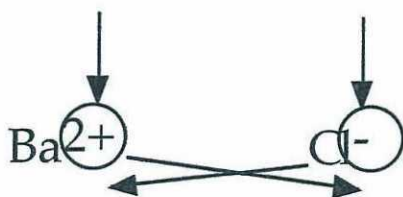


TAKE ONE OF
EACH ION.



barium chloride

NUMBERS ARE DIFFERENT.



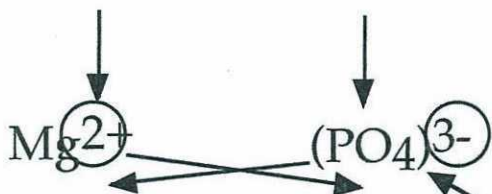
CROSS NUMBERS
DOWN.



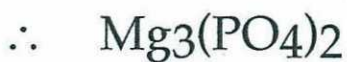
THE "1" DOESN'T HAVE
TO BE PUT IN.

magnesium phosphate

NUMBERS ARE DIFFERENT.



CROSS NUMBERS
DOWN.



USE BRACKETS FOR IONS
WITH MORE THAN ONE
ATOM PRESENT.

Note

The table of ions will be given to you for use in tests and exams - just know how to use them!

NAMING COMPOUNDS

Ionic Compounds

Ionic compounds have a *metal element* combined with a *non-metal element* (or elements).

Use the Periodic table to **identify if a metal element is present**. If there is, then the *substance is ionic*.

Rules for Naming Ionic Compounds

- Name the metal first, then the ion.
- Use the table of common ions (see separate sheet) to help name the compound.

Examples

<u>Formula</u>	<u>Name</u>	<u>Description</u>
MgCO ₃	magnesium carbonate	1xMg, 1xC, 3xO atoms
Zn(OH) ₂	zinc hydroxide	1xZn, 2xO, 2xH atoms
This multiplies everything in the bracket.		
Cu ₃ (PO ₄) ₂	copper phosphate	3xCu, 2xP, 8xO atoms

Covalent Compounds

Covalent compounds have *all non-metal atoms*.

Use the Periodic Table to *identify if a metal is present*. If *not*, then it is *most likely covalent*.

Exception - any substance with the ammonium ion (NH_4^{1+}) in it is ionic.

e.g. ammonium chloride NH_4Cl

Rules For Naming Covalent Compounds

- Use prefixes such as mono-, di-, tri-, etc.

i.e.	mono -	one	penta -	five
	di -	two	hexa -	six
	tri -	three	hepta -	seven
	tetra -	four	octa -	eight

Examples

<u>Formula</u>	<u>Name</u>	<u>Description</u>
SO ₃	sulphur trioxide	1xS, 3xO atoms
CO ₂	carbon dioxide	1xC, 2xO atoms
N ₂ O ₄	dinitrogen tetroxide	2xN, 4xO atoms
SF ₆	sulphur hexafluoride	1XS, 6xF atoms

TABLE OF COMMON IONS (A VALENCE TABLE)

Below is listed a table of common ions which MUST BE LEARNT. It is impossible to succeed at chemistry unless these symbols are committed to memory and can be recalled for instant use.

POSITIVE IONS			NEGATIVE IONS		
+1	Hydrogen	H^+	-1	Chloride	Cl^-
	Potassium	K^+		Iodide	I^-
	Silver	Ag^+		Fluoride	F^-
	Sodium	Na^+		Acetate	CH_3COO^-
	Copper I	Cu^+		Hydrogen carbonate	HCO_3^-
	Ammonium	NH_4^+		Hydroxide	OH^-
				Nitrate	NO_3^-
				Permanganate	MnO_4^-
+2	Barium	Ba^{2+}	-2	Oxide	O^{2-}
	Calcium	Ca^{2+}		Sulfide	S^{2-}
	Copper II	Cu^{2+}		Carbonate	CO_3^{2-}
	Iron II	Fe^{2+}		Sulfate	SO_4^{2-}
	Magnesium	Mg^{2+}			
	Lead	Pb^{2+}			
	Zinc	Zn^{2+}			
+3	Aluminium	Al^{3+}	-3	Phosphate	PO_4^{3-}
	Iron III	Fe^{3+}			

WHAT HAVE YOU LEARNED?

Refer to your TABLE OF COMMON IONS and try the following

- How many electrons does a hydrogen atom lose to become an ion? _____
- How many electrons does an oxygen atom gain to become ionized? _____
- Does a zinc atom gain or lose electrons to become ionized? _____
- If an atom commonly loses three electrons to become ionized. Which element could it belong to? _____
- The formula of copper II oxide is CuO . How many electrons are transferred from the copper to the oxygen atoms? _____
- Explain why *two* sodium ions are needed to combine with *one* sulfate ion to form the ionic solid sodium sulfate, Na_2SO_4 .

- Explain why *two* nitrate ions are needed to combine with *one* lead ion to form the ionic solid lead nitrate, $\text{Pb}(\text{NO}_3)_2$

- Circle the polyatomic ions Ba^{2+} , OH^- , CO_3^{2-} , Cu^+ , Fe^{3+} , SO_4^{2-} , F^- , CH_3COO^- , Ag^+ , PO_4^{3-} .

- *Without looking* at your TABLE of COMMON IONS, write in the formula of each of the following ions copper I _____, hydroxide _____, lead _____, calcium _____, iodine _____, barium _____, oxide _____, carbonate _____.

- Name these compounds

CuSO_4 _____	Al_2O_3 _____
AgCl _____	Na_2SO_4 _____
$\text{Ba}(\text{OH})_2$ _____	PbI_2 _____
NH_4NO_3 _____	KCl _____
CaCO_3 _____	NaCl _____

Now try writing formula (remember to balance the ionic charges)

potassium iodide _____
potassium permanganate _____
copper II sulfate _____
lead nitrate _____
sodium carbonate _____

WRITE THE CHEMICAL FORMULAE

COMPLETE THE TABLES BELOW BY WRITING THE CORRECT CHEMICAL FORMULA IN EACH SPACE

1	NEGATIVE IONS					
POSITIVE IONS	Cl^-	O^{2-}	OH^-	NO_3^-	SO_4^{2-}	CO_3^{2-}
Na^+						
Mg^{2+}						
Zn^{2+}						
Cu^{2+}						
Al^{3+}						
Fe^{3+}						

2	NEGATIVE IONS					
POSITIVE IONS	iodide	sulfide	hydroxide	nitrate	sulfate	phosphate
ammonium						
potassium						
calcium						
tin (II)						
lead						
iron (III)						

Chemical formulas - what a combo!

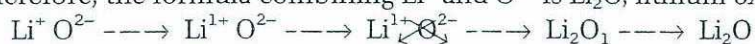
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+

Note: You will need to refer to a copy of the periodic table for this worksheet.

When you write the formula for a compound, you need to balance its charges by writing numbers as subscripts. For example, Li^+ has a one plus charge and O^{2-} has a two minus charge. When you write a little '2' after the lithium symbol, you are saying that there are two lithiums now. Since each one has a plus one charge, two of them have a two plus charge. And that balances exactly the two minus charge from O^{2-} . Therefore, the formula combining Li^+ and O^{2-} is Li_2O , lithium oxide.



1 Look at the table about chemicals below. Write down each combination in words. Then write down the chemical formula. Some have been done for you already.

	Chloride Cl^-	Oxide O^{2-}	Iodide I^-	Sulfide S^{2-}	Nitrate NO_3^-	Sulfate SO_4^{2-}	Hydroxide OH^-	Carbonate CO_3^{2-}
Lithium Li^+	Lithium chloride LiCl	Lithium oxide Li_2O	Lithium iodide LiI	Lithium sulfide Li_2S	Lithium nitrate LiNO_3	Lithium sulfate Li_2SO_4	Lithium hydroxide LiOH	Lithium carbonate Li_2CO_3
Hydrogen H^+			HI			H_2SO_4		
Magnesium Mg^{2+}			MgI_2		$\text{Mg}(\text{NO}_3)_2$			
Sodium Na^+		Na_2O			NaNO_3			
Copper Cu^{2+}				CuS				CuCO_3
Iron Fe^{3+}	FeCl_3	Fe_2O_3						
Calcium Ca^{2+}				CaS			$\text{Ca}(\text{OH})_2$	
Zinc Zn^{2+}	ZnCl_2					ZnSO_4		
Aluminium Al^{3+}			AlI_3					$\text{Al}_2(\text{CO}_3)_3$



+

2 Iron compounds can have two different formulas. For example, another formula for iron chloride is FeCl_2 . This is often called iron II chloride in order to distinguish it from the iron III chloride shown in the table. Write down the formulas of the other iron II compounds from the table.

3 Copper compounds can also have two formulas: copper I compounds and copper II compounds.

a State whether copper I or copper II compounds are in the table. _____

b Write the formulas for the other copper compounds.

4 Use the table above and your periodic table to write the formula for each of these compounds:

a lead nitrate _____

b potassium hydroxide _____

c silver bromide _____

d barium chloride _____

e potassium iodide _____

5 Name the following compounds:

a AgNO_3 _____

b K_2S _____

c BaS _____

d AgI _____

e KI _____

6 Find the following chemical names in the puzzle below. The 13 remaining letters will spell out an important chemical list.

lithium Li hydrogen magnesium Mg sodium Na copper Cu iron Fe calcium
zinc Zn aluminium chloride oxide iodide sulfide nitrate sulfate hydroxide
carbonate lead potassium silver Ag barium bromide

Answer: _____

C	H	L	O	R	I	D	E	Z	S	O	D	I	U	M
A	E	T	A	F	L	U	S	I	M	X	A	R	B	A
L	I	T	H	I	U	M	P	N	G	I	E	O	R	G
C	O	E	Y	R	L	I	D	C	I	D	L	N	O	N
I	D	E	D	I	F	L	U	S	F	E	O	D	M	E
U	I	T	R	A	L	U	M	I	N	I	U	M	I	S
M	D	C	O	P	P	E	R	E	V	L	I	S	D	I
I	E	A	G	M	U	I	S	S	A	T	O	P	E	U
C	U	A	E	H	Y	D	R	O	X	I	D	E	C	M
L	B	G	N	I	T	R	A	T	E	E	N	A	Z	N
B	A	R	I	U	M	E	T	A	N	O	B	R	A	C

WORKSHEET: CHEMICAL FORMULAE

1. *Without looking at your notes*, write the symbol and valency for the following ions. The more you can commit to memory, the easier it will be to write formulae correctly.

potassium ion

nitrate ion

magnesium ion

calcium ion

hydroxide ion

sulphate ion

aluminium ion

oxide ion

chloride ion

sodium ion

silver ion

carbonate ion

phosphate ion

sulphide ion

zinc ion

iron (III) ion

bromide ion

copper (II) ion

2. Write the formulae for the following compounds.

sodium chloride

aluminium oxide

calcium chloride

magnesium sulphate

potassium hydroxide

aluminium sulphate

potassium sulphide

copper (I) bromide

magnesium nitrate

calcium phosphate

silver oxide

iron (III) sulphide

aluminium chloride

sodium sulphate

calcium carbonate

zinc nitrate

aluminium sulphide

zinc bromide

sodium carbonate

magnesium phosphate

WRITING AND BALANCING EQUATIONS

These show the chemical change that occurs in a reaction.

Rules

1. *Write the equations in words to determine the reactants and products.*

- This step may be omitted.

2. *Write the chemical equation using the correct chemical formulas.*

3. *Subscripts are put after each compound to show the physical state the compound is in.*

(s) = solid (l) = liquid (g) = gas (aq) = in solution

e.g. $\text{H}_2\text{O}_{(\text{g})}$ = steam $\text{H}_2\text{O}_{(\text{s})}$ = ice $\text{NaCl}_{(\text{aq})}$ = salt dissolved in solution.

4. *Count the number and types of atoms on both sides of the equation.*

- Polyatomic ions (like SO_4^{2-}), which do not change from the reactants to the products side, should be considered as a group.

5. *Add numbers in front of the formulas.*

- Make sure that the total number of atoms of reactants is the same as the total number of atoms of products.
- Tick the elements off as you balance them.
- Begin with the element, other than hydrogen or oxygen, that has the greatest number of atoms in any reactant or product.
- Make sure you never change the formula of a chemical substance to balance an equation.

6. *Do a final check.*

Examples

Balance the following equations.

1. Solid zinc hydroxide + sulphuric acid solution $\rightarrow$ zinc sulphate solution + water
2. Lead (II) nitrate solution + potassium phosphate solution $\rightarrow$ solid lead (II) phosphate + potassium nitrate solution
3. heptane (C_7H_{16}) + oxygen gas $\rightarrow$ carbon dioxide gas + water vapour
4. Copper metal + nitric acid solution $\rightarrow$ copper (II) nitrate solution + nitrogen dioxide gas + water

EXERCISE 31 Balance the following equations.

1. $\text{Na} + \text{Cl}_2 \longrightarrow \text{NaCl}$
2. $\text{CO} + \text{O}_2 \longrightarrow \text{CO}_2$
3. $\text{Al} + \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3$
4. $\text{HgO} \longrightarrow \text{Hg} + \text{O}_2$
5. $\text{Mg} + \text{N}_2 \longrightarrow \text{Mg}_3\text{N}_2$
6. $\text{P}_4 + \text{O}_2 \longrightarrow \text{P}_4\text{O}_{10}$
7. $\text{Al} + \text{F}_2 \longrightarrow \text{AlF}_3$
8. $\text{PbO} + \text{C} \longrightarrow \text{Pb} + \text{CO}_2$
9. $\text{CH}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
10. $\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
11. $\text{Ca} + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{H}_2$
12. $\text{Fe} + \text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2$
13. $\text{MgCO}_3 + \text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
14. $\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
15. $\text{Mg} + \text{AgNO}_3 \longrightarrow \text{Mg(NO}_3)_2 + \text{Ag}$

EXERCISE 33 Balance the following equations.

1. $\text{H}_2\text{S} + \text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{SO}_2$
2. $\text{Al} + \text{AgNO}_3 \longrightarrow \text{Al(NO}_3)_3 + \text{Ag}$
3. $\text{NaOH} + \text{H}_3\text{PO}_4 \longrightarrow \text{Na}_3\text{PO}_4 + \text{H}_2\text{O}$
4. $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$
5. $\text{NaOH} + \text{Fe(NO}_3)_2 \longrightarrow \text{Fe(OH)}_2 + \text{NaNO}_3$
6. $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$
7. $\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu(NO}_3)_2 + \text{H}_2\text{O} + \text{NO}_2$
8. $\text{NH}_3 + \text{CuO} \longrightarrow \text{Cu} + \text{H}_2\text{O} + \text{N}_2$
9. $\text{Al}_2(\text{CO}_3)_3 + \text{HCl} \longrightarrow \text{AlCl}_3 + \text{H}_2\text{O} + \text{CO}_2$
10. $\text{FeS} + \text{O}_2 \longrightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$

ACID PROPERTIES

Acids

- Contain the *element H*.
- Produce H^{1+} ions in solution.
- Have a *sour taste*.
- Turn *blue litmus red*.

e.g. HCl - hydrochloric acid
 HNO₃ - nitric acid
 H₂SO₄ - sulphuric acid
 CH₃COOH - acetic acid

The concentration of acids (or any solution) is given by its *molarity*.

i.e. 2 M H₂SO₄ is more concentrated than 0.5 M H₂SO₄

Bases

- Any *metal oxide or hydroxide*.
- Produce OH^{1-} ions in solution.
- Have a *soapy feel* and *bitter taste*.
- Turn *red litmus blue*.

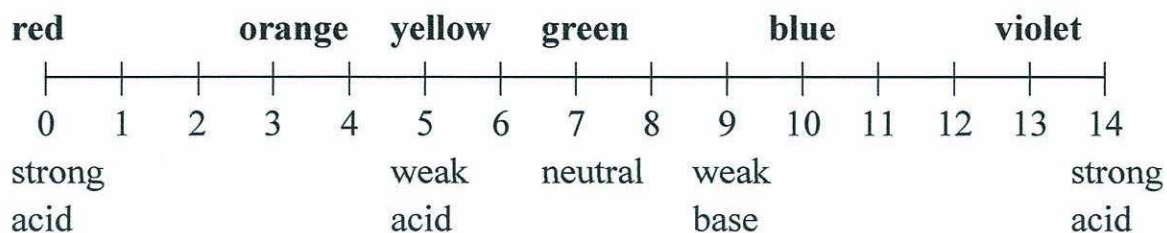
e.g. NaOH CaO
 Mg(OH)₂ NH₄OH

Note

NH₄¹⁺ behaves as if it is a metal!

pH SCALE

- Used to determine if a substance is an acid or base.



REACTIONS

When an acid reacts with another substance, it produces a *salt*.

Salt: non-metal part of an acid combined with a metal.

Examples

HCl - produces *chloride* salts.

e.g. magnesium chloride, sodium chloride, etc.

HNO₃ - produces *nitrate* salts.

e.g. magnesium nitrate, sodium nitrate, etc.

H₂SO₄ - produces *sulphate* salts.

e.g. magnesium sulphate, sodium sulphate, etc.

CH₃COOH - produces *acetate* salts.

e.g. magnesium acetate, sodium acetate, etc.

A special reaction of acids is *neutralisation*.

Neutralisation: where a base is added to an acid (or vice versa) until the solution contains neither an acid nor a base.

In the case of a large acid spill in the environment, say on a road, emergency service workers will neutralise the acid by adding enough base to neutralise it. Then it can be safely cleaned up.

ACID REACTIONS

- Only word equations are required for Year 9.

Reactions with Metals



e.g. magnesium + hydrochloric acid → magnesium chloride + hydrogen

zinc + sulphuric acid → zinc sulphate + hydrogen

NOTE - This does not apply to nitric acid. It produces another gas (NO or NO₂).

Reactions With Bases



e.g. sodium hydroxide + acetic acid → sodium acetate + water

copper oxide + hydrochloric acid → copper chloride + water

aluminium hydroxide + sulphuric acid → aluminium sulphate + water

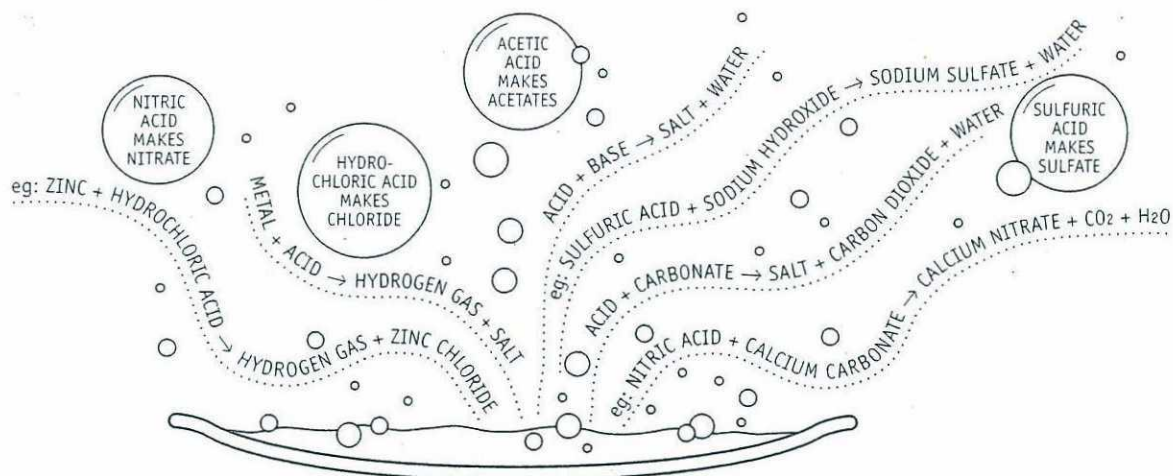
Reaction With Carbonates / Hydrogencarbonates



e.g. ammonium carbonate + nitric acid → ammonium nitrate + water + carbon dioxide

silver hydrogencarbonate + acetic acid → silver acetate + water + carbon dioxide

barium carbonate + hydrochloric acid → barium chloride + water + carbon dioxide



1 Complete the following word equations using the rules above to help you. Then draw a circle around all the acids, box the bases, underline the carbonates and highlight all the metals.

- iron + hydrochloric acid → iron chloride + _____
- magnesium + hydrochloric acid → _____ + hydrogen gas
- zinc + hydrogen sulfate → zinc sulfate + _____
- magnesium + sulfuric acid → _____ + _____
- lead + hydrogen nitrate → _____ + _____
- zinc + acetic acid → zinc acetate + _____
- potassium + acetic acid → _____ + _____
- sodium hydroxide + nitric acid → _____ + _____
- magnesium carbonate + sulfuric acid → _____ + _____ + _____
- calcium carbonate + hydrochloric acid → _____ + _____ + _____
- iron carbonate + acetic acid → _____ + _____ + _____
- _____ + _____ → hydrogen gas + sodium chloride
- _____ + _____ → sodium sulfate + water

2 Write a word equation for the following reactions, then box the reactants and circle the products for each reaction.

- Hydrochloric acid was added to calcium carbonate powder. Bubbles of carbon dioxide formed together with water. The water was evaporated leaving crystals of calcium chloride.

- Sulfuric acid was added to a solution of potassium hydroxide. After the reaction the water that formed was evaporated, leaving behind potassium sulfate.

- A piece of iron was dropped into some nitric acid. Hydrogen gas was produced leaving behind iron nitrate.

- When copper carbonate was added to hydrochloric acid, carbon dioxide and water were produced together with _____.

- A small piece of calcium was added to acetic acid producing calcium acetate and bubbles of _____.

PRECIPITATION REACTIONS

When two soluble ionic solutions are mixed it is possible that one of the new combinations of ions produces an **insoluble salt**. These types of reactions are known as **precipitation reactions**.

To predict whether a precipitate will form, a table of **solubility rules** is used.

SOLUBLE

All **Group I** cations (*lithium, sodium, potassium ions*), *ammonium ion*.

All *nitrates* and *acetates*.

All *chlorides, bromides, iodides* except *silver, lead (II) + mercury ions*.

All *sulphates* except *barium, lead (II) + calcium ions*.

INSOLUBLE

All *oxides* and *hydroxides* except **Group I** (*lithium, sodium, potassium ions*),
ammonium + barium ions.

All *sulphides* except **Group I** (*lithium, sodium, potassium ions*), *ammonium ion*.

All *carbonates, phosphates* and *sulphites* except **Group I** (*lithium, sodium, potassium ions*), *ammonium ion*.

Using The Solubility Rules

Barium nitrate and sodium sulphate solutions are mixed. Will a precipitate form?
If yes, write a molecular and net ionic equation to represent the reaction.

1. **Write down the name for each solution being mixed.** They can both be indicated as (aq) because they are soluble:

barium nitrate (aq) + sodium sulphate (aq)

2. **Look at the two possible new combinations of ions by swapping the names of the metal ions. Check out the solubility table to see if an insoluble substance is produced.**

The solubility rules indicate that, of the two possible products **barium sulphate** and **sodium nitrate**, the barium sulphate is insoluble.

3. **Write out an equation that identifies the insoluble substance formed with an (s).** The other possible combination can be represented with an (aq) to indicate that it remains in solution and is soluble.

barium nitrate (aq) + sodium sulphate (aq) → barium sulphate (s) + sodium nitrate (aq)

Examples

Write word equations for the following mixtures of solutions. If no precipitate forms, write "no reaction".

- (a) sodium phosphate + aluminium nitrate
- (b) copper (II) chloride + silver nitrate
- (c) ammonium sulphide + potassium bromide
- (d) calcium acetate + sodium carbonate
- (e) potassium hydroxide + lead (II) nitrate
- (f) calcium nitrate + sodium sulphate
- (g) sodium sulphite + potassium phosphate



+

When salts are added to water, they often dissolve forming a positive and a negative ion. When solutions of two different salts are mixed, it is possible that one of the positive ions will join with one of the negative ions to form a solid, which will not dissolve. This new solid is called a *precipitate*. Precipitates usually make the solution opaque and colourful. Eventually the precipitate settles to the bottom leaving a transparent solution on top.

Below is a table showing whether mixtures of positive ions (all metals here) and negative ions are soluble (✓) in water or are insoluble (✗) in water. If they are insoluble, they form a precipitate.

Positive ion	Negative ion				
	Carbonate	Chloride	Hydroxide	Nitrate	Sulfate
Barium	✗	✓	✓	✓	✗
Calcium	✗	✓	✓	✓	✗
Copper	✗	✓	✗	✓	✓
Lead	✗	✗	✗	✓	✗
Magnesium	✗	✓	✗	✓	✓
Potassium	✓	✓	✓	✓	✓
Silver	✗	✗	✗	✓	✗
Sodium	✓	✓	✓	✓	✓

Below is a table showing the colours of many salts.

Positive ion	Negative ion				
	Carbonate	Chloride	Hydroxide	Nitrate	Sulfate
Barium	white	white	white	white	white
Calcium	white	white	white	white	white
Copper	green	green	blue	blue	blue
Lead	white	white	white	white	white
Magnesium	white	white	white	white	white
Potassium	white	white	white	white	white
Silver	white	white	brown	white	white
Sodium	white	white	white	white	white

1 Write word equations for the reactions, when solutions of the following salts are mixed. Underline the name of the precipitate, if one forms. One example has been done for you.

a sodium carbonate + copper chloride → sodium chloride + copper carbonate

b silver nitrate + potassium chloride → _____ + _____

c copper nitrate + calcium chloride → _____ + _____

d barium nitrate + potassium hydroxide → _____ + _____

e lead nitrate + sodium sulfate → _____ + _____

2 Predict the colour of the precipitate in the following reactions. If no precipitate forms, write 'no precipitate'.

a magnesium nitrate + sodium carbonate → _____



+

- b calcium hydroxide + potassium chloride → _____
- c sodium chloride + silver nitrate → _____
- d copper chloride + potassium carbonate → _____
- e barium nitrate + copper sulfate → _____

When formulas are written for a compound which is made up of positive and negative ions, the total charge of the compound has to be zero. The superscript behind the symbol in the table shows you what the charge of the ion is. When you balance the ions in a compound, you write subscripts to indicate how many positive or negative ions there are. Here is an example worked out for you:

Ba^{2+} and Cl^- become BaCl_2
 ie the one 2+ ion of barium is balanced by two 1- ions of chloride
 $\text{Ba}^{2+} \text{Cl}^- \rightarrow \text{Ba}^{2+} \cancel{\text{Cl}^-} \rightarrow \text{Ba}_1\text{Cl}_2 \rightarrow \text{BaCl}_2$

3 Write the formulas for the possible compounds in the table below. Some examples have been done for you already.

Positive ion	Negative ion			
	Carbonate (CO_3) ²⁻	Chloride Cl^-	Hydroxide (OH) ⁻	Nitrate (NO_3) ⁻
Calcium Ca^{2+}		CaCl_2		
Copper Cu^{2+}				$\text{Cu}(\text{NO}_3)_2$
Magnesium Mg^{2+}			$\text{Mg}(\text{OH})_2$	
Silver Ag^+	Ag_2CO_3			
Sodium Na^+				

4 Answer the following questions about the molecules in the formula table above. The questions relate to *one* particle of each of the compounds.

- a How many positive ions are there in silver carbonate? _____
- b How many negative ions are there in magnesium hydroxide? _____
- c How many ions are in copper carbonate? _____
- d How many ions are there in sodium chloride? _____
- e How many atoms are in the calcium ion? _____
- f How many atoms are in the carbonate ion? _____
- g How many atoms are there in copper chloride? _____
- h How many atoms are there in silver nitrate? _____
- i Which is a molecule with the greatest number of atoms? _____
 How many atoms are there? _____
- j Which is a molecule with the smallest number of atoms? _____
 How many atoms are there? _____